

		answer
2(a)(i)	Gravitational force on satellite provides for centripetal force to keep satellite in orbit. $\frac{GM_E M_s}{r^2} = M_s r \omega^2$ $\frac{GM_E}{r^2} = r \left(\frac{2\pi}{T} \right)^2$ $T = \sqrt{\frac{4\pi^2 r^3}{GM_E}} = \sqrt{\frac{4\pi^2 (9.0 \times 10^6)^3}{(6.67 \times 10^{-11})(5.97 \times 10^{24})}}$ $= 8500s$	[1] correct substitution [1] correct answer
2(a)(ii)	Work done = increase in total energy of satellite $= E_{T,f} - E_{T,i}$ $= \left(-\frac{GM_E M_s}{2r_f} \right) - \left(-\frac{GM_E M_s}{2r_i} \right)$ $= \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(500)}{2} \left(-\frac{1}{1.0 \times 10^7} + \frac{1}{9.0 \times 10^6} \right)$ $= 1.1 \times 10^9 \text{ J}$	[1] correct substitution [1] correct answer
2(b)(i)	The gravitational force <u>acts towards centre of Earth</u> provides the centripetal force and thus, <u>gravitational force must lie on the plane of orbit which passes through centre of Earth</u>. Additionally, in order for the satellite to orbit such that it remains at a fixed relative position above the Earth, <u>its axis of rotation must coincide with Earth's axis of rotation</u> and thus it has to orbit in the equatorial plane.	[1] [1] [1]
2(b)(ii)	Geostationary satellites move in orbits where they are at a fixed relative position above the Earth. This allows the satellites to <u>constantly receive and transmit telecommunication signals to and from the same position</u> on the Earth's surface.	[1]
3(a)(i)	The internal energy of a system is the sum of a random	